

JIM MCBURNEY-LIN, PHD

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SUMMARY

- Interdisciplinary scientist with experience in neuroscience, spatial biology, and data science
- Self-motivated, problem-solving, and collaborative scientist with excellent communication skills

TECHNICAL SKILLS

- **Microscopy/Imaging:** Confocal microscopy; Epifluorescent microscopy; Automated whole slide scanning; Cell imaging and analysis
- **Molecular biology:** PCR; AAV, Retroviral transduction; immunostaining; RNAscope; in-situ hybridization; electrophysiology (32-channel tetrode, neuropixel)
- **Computational:** Programming languages (R, Python, MATLAB, SQL); Image Analysis (QuPath, ImageJ); Other (Microsoft Word, Excel, Powerpoint)

RESEARCH EXPERIENCE

Acuity Spatial Genomics

San Jose, Ca

Senior Applications DVT Engineer

December 2023 to Present

- Leading the effort to test and qualify assay automation hardware and software
- Developing novel analysis pipelines for assay, hardware, and software test routines
- Collaborating with hardware and software development teams to design new, innovative features and improvements
- Collaborating with assay development team to optimize the intersection of assay performance and hardware performance

Akoya Biosciences

Menlo Park, Ca

Scientist I

April 2022 to February 2023

Scientist II

March 2023 to December 2023

- Led a team to develop novel rapid-throughput workflows on the PhenoCyclerFusion, an automated staining and whole-slide scanning instrument
- Developed and validated image analysis routines to assess staining and imaging quality
- Designed and implemented novel automation routines to improve workflow
- Assisted in development of hardware and software improvements

University of California Riverside

Riverside, Ca

Graduate Researcher

September 2016 to March 2022

- Led a multi-disciplinary team towards characterizing the role of locus coeruleus in somatosensory-based rodent behavior
- Developed a novel animal behavioral assay to elucidate neural mechanisms of behavioral flexibility
- Designed custom behavioral apparatuses with simultaneous electrophysiology, pupil diameter tracking, and rodent movement tracking

EDUCATION

- PhD, Neuroscience, University of California Riverside, 2022
- BS, Physiology and Neuroscience, University of California San Diego, 2014

AWARDS

- Chancellor's Distinguished Fellowship (UCR, 2016)
- James and Margaret Lesley Annual Prize in Biology (UCR, 2022)

PUBLICATIONS

McBurney-Lin, J., Lu, J., Zuo, Y., Yang, H. (2019). Locus coeruleus-norepinephrine modulation of sensory processing and perception: A focused review. *Neuroscience and Biobehavioral Reviews*, 105, 190-199.

Megemont, M. *, **McBurney-Lin, J.** *, Yang, H. (2022). Pupil diameter is not an accurate real-time readout of locus coeruleus activity. *eLife*, 11, e70510.

McBurney-Lin, J. *, Sun, Y. *, Tortorelli, L.S., Nguyen, Q.A.T., Haga-Yamanaka, S., Yang, H. (2020). Bidirectional pharmacological perturbations of the noradrenergic system differentially affect tactile detection. *Neuropharmacology*, 174, 108151.

McBurney-Lin, J. *, Vargova, G. *, Garad, M., Zagha, E., Yang, H. (2022). The locus coeruleus mediates behavioral flexibility. *Cell Reports*, 41(4), 111534.

Tortorelli, L.S., **McBurney-Lin, J.**, Garad, M., Megemont, M., Riccomagno, M., Yang, H. (2022). Physiological and functional heterogeneity in the mouse locus coeruleus. *bioRxiv*, 2022.08. 19.504582.

Megemont, M., Tortorelli, L.S., **McBurney-Lin, J.**, Cohen, J. Y., O'Connor, D.H., Yang, H. (2024). Simultaneous recordings of pupil size variation and locus coeruleus activity in mice. *STAR Protocols*, 5 (1), 102785.

*denotes co-authorship